

$$Y = \{xy^q - yx^q = 1\} \subseteq \mathbb{A}^2(\mathbb{F}) \supset G \times \mu_{q+1}$$

$$\gamma: Y \rightarrow \mathbb{A}^1(\mathbb{F})$$

$$(x, y) \mapsto xy^{q^2} - yx^{q^2}$$

Prop γ is a quotient of Y by G .

Proof We already showed that $\gamma \circ g = \gamma$ for all $g \in G$.

By last time, we want to show

1. γ is surj.

2. $\gamma(P) = \gamma(Q) \Leftrightarrow gP = Q$ for some $g \in G$

3. $\exists P_0 \in Y$ s.t. $D\gamma(P_0)$ is surjective.

To show 1+2 it suffices to show that for any $a \in \mathbb{F} = \mathbb{A}^1(\mathbb{F})$,

$$|\gamma^{-1}(a)| = |G| = (q^2 - 1)q$$

$$\gamma^{-1}(a) = \{(x, y) \in \mathbb{F} \times \mathbb{F} : xy^q - yx^q = 1 \text{ and } xy^{q^2} - yx^{q^2} = a\}$$

The change of vars $(z, t) = (x, \frac{y}{x})$,

$$\gamma^{-1}(a) \xrightarrow{\sim} \mathcal{E}_a := \{(z, t) \in \mathbb{F}^x \times \mathbb{F}^x : t^q - t = \frac{1}{z^{q+1}} \text{ and } t^{q^2} - t = \frac{a}{z^{q^2+1}}\}$$

Since $t^{q^2} - t = (t^q - t)^q + (t^q - t)$, we have

$$\mathcal{E}_a = \{(z, t) \in \mathbb{F}^x \times \mathbb{F}^x : t^q - t = \frac{1}{z^{q+1}} \text{ and}$$

$$\frac{1}{z^{q^2+q}} + \frac{1}{z^{q+1}} = \frac{a}{z^{q^2+1}}\}$$

$$= \{(z, t) : \mathbb{F}^x \times \mathbb{F}^x : t^q - t = \frac{1}{z^{q+1}} \text{ and } z^{q^2-1} - az^{q-1} + 1 = 0\}$$

The poly $z^{q^2-1} - az^{q-1} + 1$ has $q^2 - 1$ distinct roots, as it is separable.
For any fixed z , $t^q - t = \frac{1}{z^{q+1}}$ has q distinct roots. So $|\mathcal{E}_a| = (q^2 - 1)q$.

Now take any $P_0 = (x_0, y_0) \in Y$. To compute the tangent space at P , work in $\mathbb{F}[\varepsilon] = \mathbb{F}[X]/(\varepsilon^2)$, consider

$$(x_0 + \varepsilon x)(y_0 + \varepsilon y)^q - (y_0 + \varepsilon y)(x_0 + \varepsilon x)^q = 1$$

$$(x_0 + \varepsilon x)y_0^q - (y_0 + \varepsilon y)x_0^q = 1$$

$$\varepsilon(y_0^q x - x_0^q y) = 0$$

$$\Rightarrow \text{Tan}_{P_0}(Y) = \{xy_0^q - yx_0^q = 0\} \subseteq \mathbb{F}^2 = \text{Tan}_{P_0}(\mathbb{A}^2(\mathbb{F}))$$

$$\text{Then } D_{P_0}\gamma: \text{Tan}_{P_0}(Y) \rightarrow \text{Tan}_{\gamma(P_0)}(\mathbb{A}^1(\mathbb{F})) = \mathbb{F}$$

$$(x, y) \mapsto (y_0^{q^2} - x_0^{q^2}) \begin{pmatrix} x \\ y \end{pmatrix} = xy_0^{q^2} - yx_0^{q^2}$$

Take $(x, y) \in \ker D_{P_0}\gamma$. Then

$$xy_0^q - yx_0^q = 0 \text{ and } xy_0^{q^2} - yx_0^{q^2} = 0$$

but

$$\det \begin{pmatrix} y_0^q & -x_0^q \\ y_0^{q^2} & -x_0^{q^2} \end{pmatrix} = -y_0^q x_0^{q^2} + x_0^q y_0^{q^2} = (x_0 y_0^q - y_0 x_0^q)^q = 1.$$

So $(x, y) = (0, 0)$ and $D_{P_0}\gamma$ is an iso. $\square$

Rmk What is γ ? Recall we have the symplectic pairing on $\mathbb{F}^2$

$$\langle (x, y), (u, v) \rangle = xv - yu$$

and it is invariant under $\mathcal{S}L_2(\mathbb{F})$.

$$\text{Then } Y = \{(x, y) : \langle (x, y), F(x, y) \rangle = 1\}$$

The map γ is $(x, y) \mapsto \langle (x, y), F^2(x, y) \rangle$

If $v = (x, y) \in Y$, then $v, F(v)$ is a unimodular basis for $\mathbb{F}_v^2$.

$$\text{So } F^2(v) = \langle F^2(v), F(v) \rangle F(v) + \langle F^2(v), v \rangle v$$

$$= F(\langle F(v), v \rangle F(v) + \langle F^2(v), v \rangle v)$$

$$= 1 \cdot F(v) + \langle F^2(v), v \rangle v$$

$$= 1 \cdot F(v) + \gamma(v) \cdot v$$

So $(x, y) = v$ satisfies

$$x^q = x^q + ax, \quad y^q = y^q + ay \quad \text{if } a = \gamma(x, y).$$

The solutions to these eqns are $\mathbb{F}_q$ -vect spaces of dim 2, V .

So (x, y) s.t. $\gamma(x, y) = a$ as ordered pair of vectors in this 2-dim vect sp. Can show $(x, y) \mapsto xy^q - yx^q$ is a nondeg alt bilin form on this 2-dim vect sp.

Then $(x, y) \in Y \Rightarrow$ det of these two vects as elements of V is a const.

$\Rightarrow \gamma^{-1}(a)$ can be describe as pairs of vectors $_{x, y}$ in $\mathbb{F}_q^2$ with $\det(x, y) = \text{fixed const.}$

Now consider

$$\phi: Y \rightarrow \mathbb{A}^1(\mathbb{F}) \setminus \{0\}$$

$$(x, y) \mapsto y$$

This satisfies

$$\bullet \text{ For } \alpha \in \mu_{q+1}, \phi(\alpha(x, y)) = \alpha \phi(y)$$

$$\bullet \phi(F(x, y)) = F\phi(x, y)$$

$$\bullet \text{ For } \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \in U, \phi\left(\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}(x, y)\right) = \phi(x + by, y) = y = \phi(x, y)$$

Prop ϕ is a quotient of Y by U .

Proof Want to satisfy 1, 2, 3 of Prop from last lecture.

For any $y \in \mathbb{A}^1(\mathbb{F}) \setminus \{0\}$, we can solve

$$xy^q - yx^q = 1 \text{ for } x,$$

so ϕ is surjective.

Take $(x, y), (x', y') \in Y$ with $\phi(x, y) = \phi(x', y')$ i.e. $y = y'$.

$$\text{Then } xy^q - yx^q = x'y^q - yx'^q$$

$$\Rightarrow \left(\frac{x}{y}\right)^q - \frac{x}{y} = \left(\frac{x'}{y}\right)^q - \frac{x'}{y}$$

$$\Rightarrow \left(\frac{x-x'}{y}\right)^q = \frac{x-x'}{y} \Rightarrow b := \frac{x-x'}{y} \in \mathbb{F}_q$$

$$\text{and } \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

So $\phi(x, y) = \phi(x', y') \Leftrightarrow (x', y') = g(x, y)$ for some $g \in U$.

Finally $D_P \phi = \begin{pmatrix} 0 & 1 \end{pmatrix}$ at any $P \in \mathbb{A}^2(\mathbb{F})$, and if

$P = (x_0, y_0) \in Y$, then $\text{Tan}_P(Y) = \{xy_0^q - yx_0^q = 0\}$.

Since $x_0, y_0 \neq 0$, $D_P \phi: \text{Tan}_P(Y) \rightarrow \text{Tan}_{\phi(P)}(\mathbb{A}^1(\mathbb{F}) \setminus \{0\})$ is $\neq 0$,
hence an iso. $\square$

Next time

$$Y/\mu_{q+1} \cong \mathbb{P}^1(\mathbb{F}) \setminus \mathbb{P}^1(\mathbb{F}_q)$$